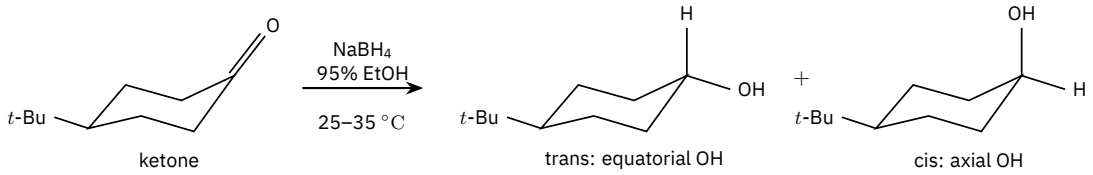
ORGANIC EXPERIMENT 3 | PRE-LAB CHECKLIST

Stereoselective reduction

4-tert-Butylcyclohexanone ⟶ cis/trans-4-tert-butylcyclohexanol

Based on the CM3191 Organic Lab Manual AY26/27, printed pp. 14–18 [M], and lecture CM3191 Org L3, slides 4–8 [L]. Layout follows the Experiment 1 and 2 bench sheets. Added practical notes or general expectations are marked [E].



ketone trans: equatorial OH cis: axial OH

1. Reagents and quantities

Reagent / product	MW (g mol^{-1})	Amount	mmol	Equiv.
4-tert-Butylcyclohexanone	154.25	0.500 g	3.24	1.00
Sodium borohydride	37.83	0.400 g	10.57	3.26
95% Ethanol	46.07	5 mL	—	solvent
4-tert-Butylcyclohexanol (total)	156.27	0.507 g (theor.)	3.24	—

Calculated from [M]: $n_{\text{ketone}} = 1000 m_{\text{ketone}}/154.25$ in mmol (mass in g). The 3.21 mmol shown on [L, slide 4] does not match 0.500 g; this sheet uses 3.24 mmol. Equivalents above count $NaBH_4$ molecules.

$$m_{\text{theor}} = m_{\text{ketone}} \frac{156.27}{154.25} = 1.0131 m_{\text{ketone}}, \qquad \text{yield} = \frac{m_{\text{dry product}}}{m_{\text{theor}}} \times 100\%.$$

Also prepare [M]: water 3 mL for quench; 3 M HCl 1 mL; EtOAc 2×10 mL; water 2×10 mL for washing; brine 10 mL; anhydrous Na_2SO_4; hexane/EtOAc 4:1 for TLC; staff-prepared anisaldehyde stain; $CDCl_3$ for NMR.

2. Bring and prepare

- ☐ Official datasheet: last page, printed
- ☐ Lab coat, goggles, gloves, covered shoes
- ☐ 25 mL rbf, stir bar, water bath, thermometer [E]
- ☐ Phone for photos of balance readings
- ☐ Ruler, pencil, marker; labelled product bag
- ☐ Pre-weighed 100 mL rbf; separation/filtration setup

Confirm with the GTA [E]: available TLC standard and sampling method; local waste streams; how to remove any inorganic solid before extraction. The manual does not specify a boiling duration or drying-agent mass. Record the actual procedure.

3. Session sequence

Reaction	Dissolve; add $NaBH_4$ slowly; maintain 25–35 °C. TLC endpoint; usually within 1 h [M].
Quench	Water, then HCl dropwise. About 10 min; gas must become very slow or cease before heating [M].
Isolation	Water-bath heating; cool; extract; wash; dry; filter; rotary evaporate; weigh [M].
Finish	IR; NMR sample; product submission; photo check and signed hardcopy datasheet [M].

SAFETY AT THE BENCH [M; practical reminders E]

Work in the fume cupboard. $NaBH_4$ and HCl are corrosive; the quench releases flammable H_2. Ethanol, EtOAc and hexane are flammable. Anisaldehyde stain is corrosive and flammable; use the staff-prepared solution. Avoid exposure to $CDCl_3$.
Keep gas-evolving vessels vented; keep ignition sources away [E]. Do not heat during vigorous gas evolution. Use a water bath, not an ice bath, to control the reaction temperature [M].

BENCH MANUAL | ENGLISH

Reaction, work-up and analysis

Tick each step. Amounts and operations follow [M, pp. 15–17]. Added checks are marked [E].

1. Reduction and TLC

☐ **1.** Dissolve **0.500 g 4-tert-butylcyclohexanone** in **5 mL 95% ethanol** in a **25 mL round-bottom flask** with a stir bar. Stir properly. Photograph the actual mass.

☐ **2.** Weigh **0.400 g $NaBH_4$**; photograph. Add it **slowly, in small portions** while stirring. Keep the mixture at **25–35** $^{\circ}C$. If warm, cool in a **water bath, not an ice bath**. Keep the flask vented [E].

☐ **3.** Monitor by TLC: **hexane/EtOAc 4:1 (v/v)**. Use ketone reference K and reaction sample R; a co-spot K+R helps identification [E]. Record sampling time and temperature.

☐ **4.** View under **UV first**; circle visible spots in pencil. Dip the **same plate** in staff-prepared **anisaldehyde stain**; remove and heat on a hotplate until coloured spots appear. Record both views.

☐ **5.** Continue until TLC indicates completion; the manual gives **typically within 1 h**. Do not use elapsed time or a blank UV view alone as the endpoint [E]. If ketone remains, consult the GTA before changing the procedure [E].

2. Controlled quench and heating

☐ **6.** Once complete, carefully add **3 mL water**. Allow for gas evolution. Keep stirring and control the addition [E].

☐ **7.** Add **1 mL 3 M HCl dropwise**, cautiously. **Hydrogen may evolve vigorously.** Keep the vessel vented and control the addition [E].

☐ **8.** After **about 10 min**, when gas evolution is **very slow or has ceased**, heat the mixture **on a water bath to boiling**. Then cool to **room temperature**. Record actual times; no boiling duration is specified.

☐ **9.** A white inorganic solid may form. It can be discarded before extraction [M]; ask the GTA how to separate it if present [E]. Do not assume every white solid is waste [E].

3. Extraction, drying and isolation

☐ **10.** Extract with **EtOAc 10 mL, twice**. Keep and combine the organic layers. Vent the separatory funnel frequently [E]. EtOAc is normally the upper layer; verify layer identity [E].

☐ **11.** Wash the **combined organic layer** with **water 10 mL, twice**, then **brine 10 mL, once**. Keep the organic layer after each wash. Retain separated layers until recovery is confirmed [E].

☐ **12.** Dry the organic layer over **anhydrous Na_2SO_4**. Record the amount used [E]. No amount or drying time is specified in the manual.

☐ **13.** Weigh a clean, dry **100 mL round-bottom flask**; photograph. Filter the dried solution into this **pre-weighed flask**, retaining the solution and removing the drying agent.

☐ **14.** Remove solvent on the **rotary evaporator**. Record the bath/pressure settings [E]. Weigh the flask with product and photograph. Product mass = (flask + product) − empty flask.

☐ **15.** Transfer the product to a **labelled plastic bag for submission**. Record appearance and calculate the total isolated yield. Residual solvent inflates the apparent mass [E].

4. Characterisation and sign-off

☐ **16.** Record the **IR spectrum**. Save the file; compare the O–H and C=O regions with the reaction/TLC evidence [L].

☐ **17.** For ^1H NMR, dissolve **10–20 mg product in 0.6 mL $CDCl_3$** in a **clean, dry test tube**. Transfer with a clean, dry glass dropper into a clean, dry NMR tube.

☐ **18.** Cap and label the NMR tube. Check for **4–6 cm solution height**; top up if needed. Submit the sample and record tube/sample identifiers.

☐ **19.** Copy actual masses and observations onto the **official datasheet**. Draw the TLC plate and label **all distances in cm**.

☐ **20.** Show measurement photos to the GTA and obtain a **signature on the printed datasheet**. Submit the signed sheet with the report. Keep the working log and raw spectra.

Keep these distinctions clear
TLC checks remaining ketone; NMR gives the cis/trans ratio. The manual specifies no recrystallisation or melting-point measurement for this experiment.

실험 매뉴얼 | 한국어

반응, 후처리 및 분석

영문 절차와 같은 순서입니다. [M] 공식 매뉴얼, [L] 강의자료, [E] 추가 설명·실무 확인 사항.

1. 환원 반응과 TLC

☐ **1.** **25 mL 둥근바닥 플라스크**에 교반 막대를 넣습니다. **4-tert-butylcyclohexanone 0.500 g**을 **95% 에탄올 5 mL**에 녹이고 잘 교반합니다. 실제 질량을 촬영합니다.

☐ **2.** **$NaBH_4$ 0.400 g**을 재고 촬영합니다. 교반하면서 **소량씩 천천히** 넣습니다. 반응액을 **25--35** °C로 유지합니다. 따뜻해지면 **얼음 수조가 아닌 물 수조**로 식힙니다. 밀폐하지 않습니다 [E].

☐ **3.** **hexane/EtOAc 4:1 (v/v)**로 TLC를 전개합니다. 출발물질 K와 반응액 R을 찍습니다. K+R co-spot을 추가하면 비교하기 쉽습니다 [E]. 채취 시각과 온도를 기록합니다.

☐ **4.** 먼저 **UV**로 확인하고 보이는 점을 연필로 표시합니다. **같은 판**을 실험실에서 준비한 **anisaldehyde stain**에 담갔다 꺼냅니다. 핫플레이트에서 가열해 색을 발현시키고, 두 관찰 결과를 기록합니다.

☐ **5.** TLC로 반응 종료를 판단합니다. 매뉴얼상 **보통 1시간 이내**입니다. 시간 경과나 UV에서 점이 안 보인다는 이유만으로 끝내지 않습니다 [E]. 케톤이 남으면 GTA와 상의합니다 [E].

2. Quench와 가열

☐ **6.** 반응이 끝나면 **물 3 mL**를 조심스럽게 넣습니다. 기체 발생에 주의하고, 교반하면서 첨가 속도를 조절합니다 [E].

☐ **7.** **3 M HCl 1 mL**를 **한 방울씩** 조심스럽게 넣습니다. **수소가 격렬하게 발생할 수 있습니다.** 밀폐하지 말고 첨가 속도를 조절합니다 [E].

☐ **8.** **약 10분 뒤**, 기체 발생이 **매우 느려지거나 멈추면 물 수조에서 혼합물이 끓을 때까지** 가열합니다. 이어서 **실온**으로 식힙니다. 실제 시간을 기록합니다. 끓이는 시간은 별도 지정되어 있지 않습니다.

☐ **9.** 흰색 무기 고체가 생길 수 있으며, 추출 전에 제거할 수 있습니다 [M]. 생겼다면 분리 방법을 GTA에게 확인합니다 [E]. 확인 없이 흰 고체를 모두 버리지 않습니다 [E].

3. 추출, 건조 및 분리

☐ **10.** **EtOAc 10 mL씩 두 번** 추출하고 유기층을 합칩니다. 분액깔때기는 자주 압력을 빼 줍니다 [E]. EtOAc 층은 보통 위층이지만, 층의 정체를 확인합니다 [E].

☐ **11.** 합친 유기층을 **물 10 mL씩 두 번**, 이어서 **brine 10 mL로 한 번** 씻습니다. 매번 유기층을 보관합니다. 회수가 확인될 때까지 분리한 층을 남겨 둡니다 [E].

☐ **12.** 유기층을 **무수 Na_2SO_4**로 건조합니다. 사용량을 기록합니다 [E]. 매뉴얼은 건조제의 양이나 건조 시간을 지정하지 않습니다.

☐ **13.** 깨끗하고 건조한 **100 mL 둥근바닥 플라스크**의 질량을 재고 촬영합니다. 건조제를 걸러 내면서 용액을 이 **미리 무게를 잰 플라스크**에 받습니다.

☐ **14.** **회전증발기**로 용매를 제거합니다. 수조 온도와 압력을 기록합니다 [E]. 플라스크와 생성물의 질량을 재고 촬영합니다. 생성물 질량 = (플라스크 + 생성물) − 빈 플라스크.

☐ **15.** 생성물을 **이름을 적은 비닐봉투**에 옮겨 제출합니다. 외관을 기록하고 전체 분리 수율을 계산합니다. 잔류 용매는 겉보기 질량을 높입니다 [E].

4. 분석과 서명

☐ **16.** **IR 스펙트럼**을 측정하고 파일을 저장합니다. O--H 및 C=O 영역을 TLC 결과와 함께 해석합니다 [L].

☐ **17.** **생성물 10--20 mg**을 깨끗하고 건조한 시험관에서 **$CDCl_3$ 0.6 mL**에 녹입니다. 깨끗하고 건조한 유리 스포이트로 NMR 튜브에 옮깁니다.

☐ **18.** NMR 튜브를 막고 라벨을 붙입니다. 용액 높이를 **4--6 cm**로 맞추고, 필요하면 용매를 보충합니다. 시료를 제출하고 튜브·시료 번호를 기록합니다.

☐ **19.** 실제 질량과 관찰을 **공식 datasheet**에 옮깁니다. TLC 판을 그리고 **모든 거리를 cm 단위**로 표시합니다.

☐ **20.** GTA에게 측정 사진을 보여 주고 **인쇄한 datasheet에 서명**을 받습니다. 서명한 원본을 보고서와 함께 제출합니다. 작업 기록과 원시 스펙트럼도 보관합니다.

분석 목적을 구분하세요
TLC는 출발 케톤의 잔류 여부, NMR은 cis/trans 비율을 확인합니다. 이번 실험의 공식 절차에는 재결정과 녹는점 측정이 없습니다.

DATASHEET | WORKING LOG

Record what actually happens

For your own notes. The original official datasheet is the last page; it must be printed and signed.

Name: KIM DONGHWAN **Group:** _____ **Fumehood:** _____ **Date:** _____

1. Masses and sample details

Ketone mass (g):	$NaBH_4$ mass (g):
Ketone: $n = 1000m/154.25$ (mmol):	$NaBH_4$: $n = 1000m/37.83$ (mmol):
Empty 100 mL rbf (g):	Rbf + dry product (g):
Product mass, by difference (g):	Theoretical mass, $1.0131m_{\text{ketone}}$ (g):
Isolated yield (%):	Product appearance / bag label:
IR file / sample ID:	NMR tube / sample ID:
NMR mass (mg) / solvent volume (mL):	^{1}H frequency (MHz; from acquisition data):

2. Times, temperature and work-up

$NaBH_4$ addition start / end	TLC sample times	Temperature range ($^\circ$C)
Water / HCl addition times	Gas very slow / stopped	Boiling / cooled to RT
EtOAc extractions (actual mL)	Water / brine washes (mL)	Na_2SO_4 / drying time
Rotavap bath / pressure	Solvent removal finished	Photo file numbers

3. TLC measurements

Eluent: hexane/EtOAc 4:1. Measure baseline → spot centre and baseline → front. Record each spot separately. Photograph before and after staining; draw the plate on the official sheet.

Plate / time	Lane	Spot (cm)	Front (cm)	R_f	UV / stain colour
	K				
	R				
	K+R				

4. Observations and departures from the procedure

Addition/exotherm; gas evolution; white solid; layer identification or emulsion; drying; final product; material losses; GTA instructions. Expectations are not results.

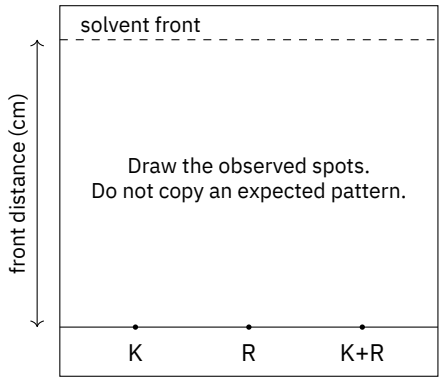
☐ Mass photos saved ☐ TLC recorded ☐ IR/NMR IDs saved ☐ Product submitted ☐ Official sheet signed

APPENDIX A | TLC AND EXTRACTION

Two checks at the bench

Visual aids [E] for the procedure in [M]. No spot heights or numerical R_f values are predicted.

1. TLC: UV first, then stain the same plate



baseline about 1 cm above the bottom [E]

K = ketone; R = reaction sample.
K+R is a co-spot of both on one origin [E]. Use a GTA-approved ketone reference and sampling method.

1. Develop in **hexane/EtOAc 4:1**. Keep the liquid level below the baseline [E].
2. Mark the front immediately [E]. View under UV and circle visible spots [M].
3. Stain the **same plate** with anisaldehyde, then heat as directed [M].
4. Compare R with K **after staining**. Photograph and record distances [E].

Why staining matters [E]: these compounds lack a strong extended UV chromophore. Weak or absent UV spots do not prove completion. On silica, the alcohol commonly runs below the ketone, but the reference comparison is essential.
A single product spot does not establish a single diastereomer. The cis/trans alcohols may be unresolved [E].

$$R_f = \frac{\text{distance from baseline to spot centre}}{\text{distance from baseline to solvent front}}$$

2. Follow the product through the work-up

Cooled, quenched reaction mixture
Extract with EtOAc 10 mL, then a fresh 10 mL [M].

↓

Combine the two EtOAc layers; keep the organic phase
Normally the upper layer. Verify rather than relying only on position [E].

↓

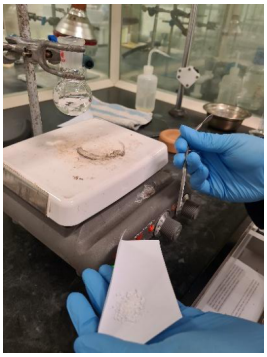
Wash the combined organic phase
Water 10 mL twice; brine 10 mL once. Keep the organic phase each time [M].

↓

Dry; filter into the pre-weighed 100 mL rbf
Anhydrous Na_2SO_4; retain the filtrate [M].

↓

Rotary evaporate; weigh; transfer to the labelled bag
Use the same flask tare to calculate the isolated mass [M].



Reaction setup during solid addition. Photo cropped from [M, printed p. 15]; original colour retained.

Add $NaBH_4$ slowly. Maintain 25–35 °C; cool with a water bath if needed [M].

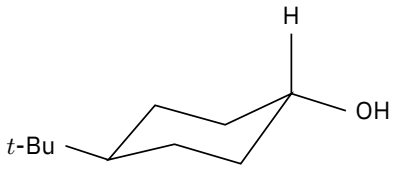
Extraction is not washing
During extraction, product moves from the quenched mixture into EtOAc. During washing, aqueous washes remove water-soluble contaminants while the product is retained in EtOAc. Keep the separated layers until product recovery is confirmed [E].

APPENDIX B | STEREOCHEMISTRY AND ANALYSIS

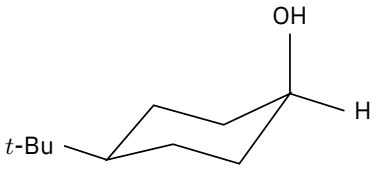
Read the carbinol proton

Use the ^{1}H attached to C–OH, not the exchangeable OH proton, for the cis/trans ratio.

The bulky t-Bu group strongly favours an equatorial position [L, slide 6]. In that dominant chair, equatorial OH is trans to t-Bu; axial OH is cis. The diagrams and approximate shifts below follow [L, slides 7–8].



trans alcohol
equatorial OH; axial C–H
$\delta_{\mathrm{H}} \approx 3.5$ ppm



cis alcohol
axial OH; equatorial C–H
$\delta_{\mathrm{H}} \approx 4.0$ ppm

An axial hydride approach gives axial C–H / equatorial OH; an equatorial approach gives equatorial C–H / axial OH [L]. Measure which product is major. The sources do not provide an expected numerical cis/trans ratio for this experiment.

1. Integration and product ratio

Assignment	Guide shift [L]	Observed range (ppm)	Integral
trans, axial C–H (1 H)	about 3.5 ppm		
cis, equatorial C–H (1 H)	about 4.0 ppm		

$$\frac{n_{\mathrm{trans}}}{n_{\mathrm{cis}}} = \frac{I_{\mathrm{trans}}}{I_{\mathrm{cis}}}, \qquad \%\mathrm{trans} = \frac{100 I_{\mathrm{trans}}}{I_{\mathrm{trans}} + I_{\mathrm{cis}}}, \qquad \%\mathrm{cis} = \frac{100 I_{\mathrm{cis}}}{I_{\mathrm{trans}} + I_{\mathrm{cis}}}.$$

Each assigned signal represents one carbinol proton. Integrate the whole multiplet using a consistent baseline. Check for solvent/impurity overlap. If the signals are not resolved, report that limitation rather than inventing a ratio [E]. Other ring signals overlap [L].

Record: trans:cis = _____ major product = _____ trans % = _____ cis % = _____

2. Combine TLC, IR and NMR evidence

Method	Look for	Do not conclude from this alone
TLC	Remaining ketone at the reference position; compare the stained plate.	No UV spot does not prove completion; one spot does not prove one isomer.
IR	Alcohol O–H; reduction of ketone C=O signal. General guides [E]: O–H 3200–3600 cm^{-1}; C=O near 1715 cm^{-1}.	An O–H band alone does not prove pure product or complete conversion.
^{1}H NMR	Carbinol-proton assignments and integrals; residual solvent or other impurities.	Approximate shifts are assignment guides, not measured results.

3. What to retain for the report

Actual reagent masses and temperature history; endpoint TLC and distances; quench observations; extraction/drying details; mass calculation and solvent-related uncertainty; IR assignments; NMR acquisition frequency, shifts and integrals; measured cis/trans ratio; justified limitations. This follows the supplied Organic Teaching Lab Report Format, pp. 1–3.

Official datasheet follows
The next page is the **unchanged official datasheet**: printed p. 18 of [M] (PDF page 19). Print it, complete it, show the measurement photos to the GTA and obtain a signature on the hardcopy. The working log on p. 4 does not replace it.

Sources and provenance. [M] Hoang, T. G. *CM3191 Organic Lab Manual AY26/27*, September 2026, printed pp. 14–18. [L] *CM3191 Org L3: Stereoselective Reduction of 4-tert-Butylcyclohexanone*, slides 4–8. [E] Added explanatory/practical notes; these do not change the prescribed reagent quantities. Reaction setup photo: cropped from [M], printed p. 15, without changing the aspect ratio. Formatting references: supplied Experiment 1 Suzuki Bench Sheets and Experiment 2 Aldol–DA Bench Sheets + Datasheet. All working fields are intentionally blank.

Name: _____

Lab Group_____ Fumehood _____ Date of the experiment: _____

Stereoselective Reduction of 4-tert-Butylcyclohexanone

Instruction: Take photos of the measurements. After completing the experiment, show the photos to your teaching assistant and ask them to sign your printed datasheet (hardcopy, not softcopy). Submit the signed datasheet along with your lab report

Starting materials

Mass of 4-*tert*-butylcyclohexanone added (g) _____

Mass of sodium borohydride added (g) _____

Product

Mass of the empty rbf (g)_____

Mass of product + rbf (g) _____

Mass of your final product (g) _____

Observations

Draw your TLC plate here and label all distances (in cm)